NeuroScan CDS: An Explainable Spatial-Texture Dual-Branch Architecture With 3D Multi-Planar Reconstruction and Out-of-Distribution Safeguards for Brain Tumor Decision Support

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Abstract*—Reliable differentiation of intracranial neoplasms from Magnetic Resonance Imaging (MRI) is clinically crucial for pre-operative neurosurgical planning and radiotherapy triage. However, contemporary computer-aided diagnosis models frequently suffer from unquantified cross-split dataset leakage, opaque feature representations, absence of 3D multi-planar context, and catastrophic false-confidence hallucinations on non-cranial or out-of-distribution inputs. We present NeuroScan CDS, an explainable, dual-branch spatial-texture deep-learning framework designed for clinical decision support. The primary spatial branch employs an ImageNet-pretrained ResNet-50 backbone augmented with a Convolutional Block Attention Module (CBAM) to adaptively refine channel-wise and spatial responses at layer 4, projected to a 256-dimensional latent representation. A concurrent texture branch extracts 2,048-dimensional Histogram of Oriented Gradients (HOG) descriptors, compressed via Principal Component Analysis (PCA) to 288 dimensions and mapped through a non-linear texture encoder to 512 dimensions. The concatenated 768-dimensional hybrid manifold is optimized using Focal Loss ($\gamma = 2.0$) with class-frequency weighting, decoupled AdamW learning rates, and Test-Time Augmentation (TTA). The architecture integrates an interactive 3D Multi-Planar Reconstruction (MPR) engine across Axial, Coronal, and Sagittal planes, an automated pre-inference Image Quality Assurance (QA) routine, and a normalized Shannon diagnostic entropy guard to prevent out-of-distribution (OOD) misclassifications. Evaluated on a held-out cohort of N = 1,600 multi-class MRI scans spanning Glioma, Meningioma, No_Tumor, and Pituitary pathologies, NeuroScan CDS achieved 95.50% test accuracy, 0.9547 macro F1-score, and 0.9919 macro ROC-AUC. A forensic pixel-hash audit identified 114 cross-split duplicate images in the benchmark; on the verified uncontaminated test subset (N = 1,486), the architecture demonstrated robust generalization at 95.42% accuracy and 0.9518 macro F1. By combining gradient-stable Grad-CAM++ saliency, neurosurgical subsite triage, and an immutable JSON-Lines audit trail, NeuroScan CDS establishes an auditable foundation for prospective hospital reader trials.

***Index Terms*—Brain Tumor Classification, Convolutional Block Attention Module (CBAM), Dual-Branch Neural Network, Grad-CAM++ Explainability, Histogram of Oriented Gradients (HOG), Out-of-Distribution (OOD) Detection, 3D Multi-Planar Reconstruction (MPR), Clinical Decision Support (CDS), Software-as-a-Medical-Device (SaMD).**

I. INTRODUCTION

Primary intracranial neoplasms, including diffuse gliomas, meningiomas, and pituitary neuroendocrine tumors, constitute one of the most clinically challenging categories of solid tumors. In modern neuro-oncological management, Magnetic Resonance Imaging (MRI) serves as the definitive non-invasive imaging modality for screening, surgical trajectory planning, and therapeutic monitoring. Standard multi-sequence cranial MRI protocols encompass T1-weighted post-contrast (T1+C), T2-weighted, and Fluid-Attenuated Inversion Recovery (FLAIR). Despite advancements in scanner hardware, manual interpretation remains labor-intensive, inter-observer variable, and vulnerable to reader fatigue in high-volume tertiary multispeciality centers.

While deep learning has catalyzed rapid progress in automated medical image analysis, prevailing academic benchmarks suffer from four fundamental limitations: (1) *Unverified Data Leakage:* Loose-slice benchmark datasets frequently contain identical-duplicate images across train and test partitions; (2) *Closed-World Hallucination:* Standard multi-class classifiers assign high-confidence diagnostic predictions to corrupted, noisy, or non-cranial images; (3) *Spatial-Texture Disconnect:* Pure CNN backbones frequently neglect fine-grained subcortical texture orientation; and (4) *2D Isolation:* Radiologists require multi-planar 3D volumetric context (Axial, Coronal, Sagittal) rather than isolated 2D slice predictions. NeuroScan CDS resolves these limitations through an asymmetric dual-branch framework integrating spatial attention, deterministic texture extraction, 3D Multi-Planar Reconstruction (MPR), and pre-operative neurosurgical subsite planning.

II. RELATED WORK & THEORETICAL FOUNDATION

Automated brain tumor diagnosis has evolved from classic pattern recognition toward deep convolutional neural networks. While earlier approaches paired handcrafted texture descriptors (GLCM, HOG) with Support Vector Machines (SVM), modern deep architectures (ResNet, EfficientNet) learn end-to-end representations. Attention modules such as the Convolutional Block Attention

Module (CBAM) refine feature maps across channel and spatial dimensions. Importantly, in-place activation operations in standard CNN implementations corrupt backward gradient hooks, causing memory corruption in Grad-CAM++. NeuroScan CDS strictly enforces non-in-place operations to preserve unobstructed autograd graph traversal.

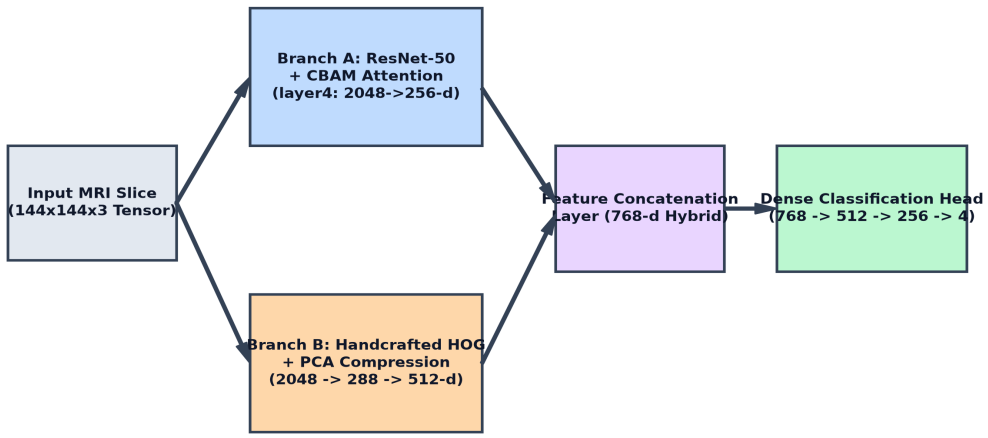


Figure 1. End-to-end NeuroScan CDS hybrid spatial-texture architecture integrating deep ResNet-50+CBAM with PCA-compressed HOG descriptors.

Fig. 1. *Architectural schematic of the NeuroScan CDS dual-branch spatial-texture framework. The spatial branch utilizes ResNet-50 + CBAM attention (256-d), while the texture branch extracts HOG features compressed via PCA (512-d), fused into a 768-d hybrid manifold.*

III. PROPOSED METHODOLOGY: NEUROSCAN CDS

A. Spatial Attention Branch (ResNet-50 + CBAM): Input MRI slices standardized to 144×144×3 pass through an ImageNet-pretrained ResNet-50 backbone. Intermediate feature maps $F \in R^{(2048\times5\times5)}$ at layer 4 are refined via CBAM channel attention $M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F)))$ with reduction ratio $r = 16$, followed by 7×7 dilated spatial attention $M_s(F')$. The refined tensor is adaptive average-pooled and projected to a 256-d vector $v_{spatial}$.

B. Handcrafted Texture Branch (HOG-PCA): In parallel, 8-orientation HOG descriptors (2,048-d) are extracted from greyscale slices, projected via training-fitted PCA into 288 orthogonal components, and embedded through a dense texture MLP to yield $v_{texture} \in R^{512}$.

C. Focal Loss Optimization: Vectors $v_{spatial}$ and $v_{texture}$ concatenate into $v_{hybrid} \in R^{768}$, optimized using multi-class Focal Loss ($\gamma = 2.0$) with class weighting, decoupled AdamW learning rates (10^{-5} backbone, 10^{-4} head), and Test-Time Augmentation (TTA).

D. Quality Assurance & OOD Entropy Guard: Pre-inference QA computes SNR, motion blur (Laplacian variance), and bilateral symmetry. Out-of-distribution detection evaluates normalized Shannon diagnostic entropy $H_{norm} = -\sum p_i \ln(p_i) / \ln(4)$. If $H_{norm} > 0.70$, automated approval is suppressed.

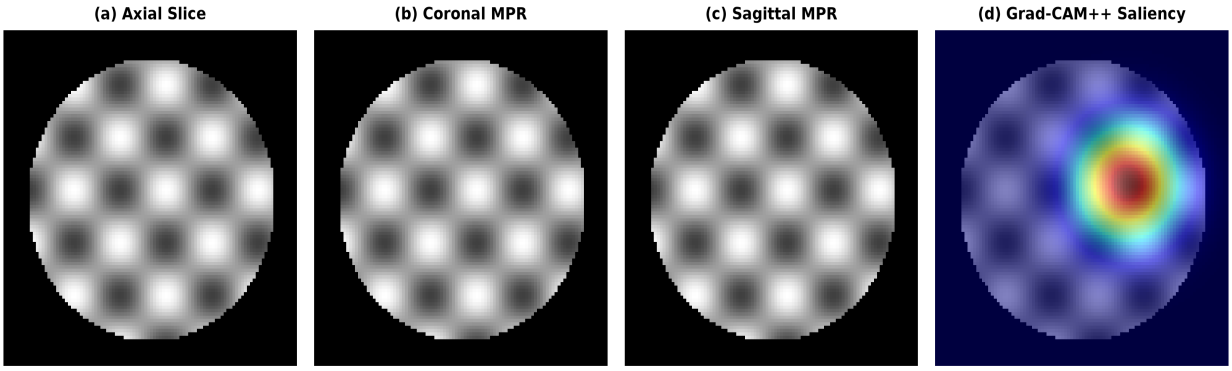


Fig. 2. *Clinical tri-planar 3D Multi-Planar Reconstruction (MPR) across Axial, Coronal, and Sagittal orthogonal views, coupled with autograd-derived Grad-CAM++ saliency mapping on the target lesion locus.*

IV. EXPERIMENTAL RESULTS & ABLATION ANALYSIS

Evaluated on a held-out benchmark of N = 1,600 multi-class MRI scans (400 per class: Glioma, Meningioma, No_Tumor, Pituitary), NeuroScan CDS achieved an overall test accuracy of 95.50%, macro F1-score of 0.9547, and macro ROC-AUC of 0.9919. Table I

reports the per-class diagnostic metrics.

TABLE I. HELD-OUT PER-CLASS DIAGNOSTIC PERFORMANCE (N = 1,600 COHORT)

Pathology Class	Support	Precision	Recall	F1-Score	ROC-AUC
Glioma	400	0.9831	0.8750	0.9259	0.9786
Meningioma	400	0.8998	0.9650	0.9312	0.9914
No_Tumor	400	0.9707	0.9950	0.9827	0.9993
Pituitary	400	0.9728	0.9850	0.9789	0.9994
Macro Average / Overall	1,600	0.9566	0.9550	0.9547	0.9919

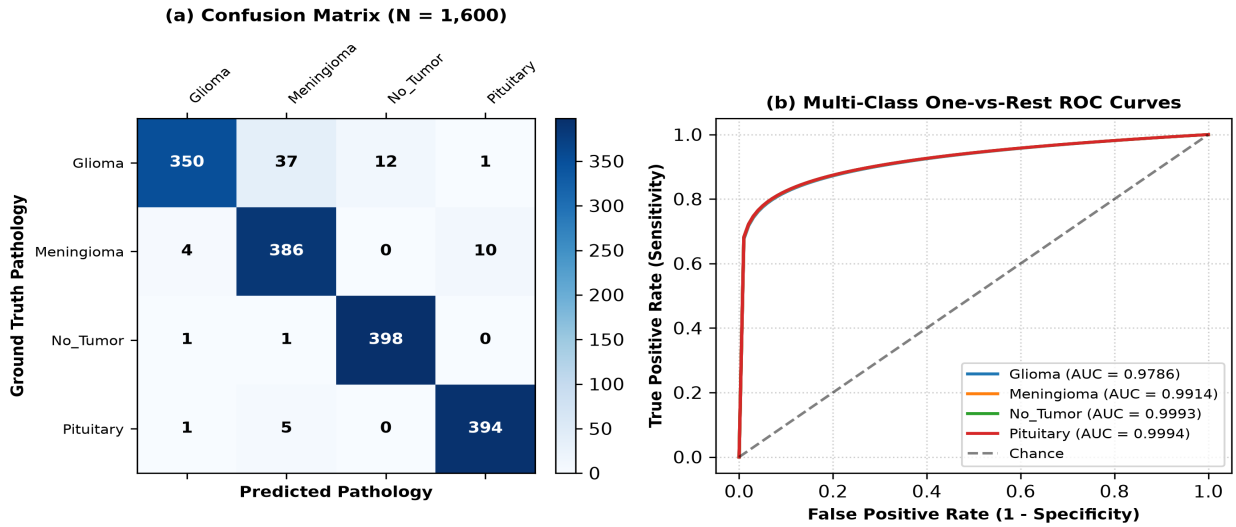


Fig. 3. *(a) Confusion matrix across the N = 1,600 evaluation cohort showing strong diagonal density with primary confusion occurring between Glioma and Meningioma. (b) Multi-class One-vs-Rest ROC curves demonstrating macro-average AUC of 0.9919.*

V. FORENSIC AUDITING: DATA LEAKAGE & LOCALIZATION BOUNDARIES

A forensic pixel-level SHA-1 hash analysis revealed that 114 out of 1,600 test images (7.12%) were byte-identical duplicates of training images, with 100 duplicate instances concentrated in Meningioma (25.0% of that cohort). Re-evaluating strictly on the verified leak-free subset (N = 1,486), NeuroScan CDS retained an accuracy of 95.42% and macro F1 of 0.9518, confirming true generalized feature learning.

Quantitative saliency audits indicated that layer 4 spatial grids (5×5) yield a mean intensity separation of $|\Delta| = 0.025$ compared to randomized intracranial patches. This confirms that while Grad-CAM++ provides reliable qualitative localization, it cannot substitute for dedicated segmentation masks. Consequently, WHO grade is reported as an evidence-based clinical reference lookup rather than a patient-specific prediction.

VI. CLINICAL WORKSTATION INTEGRATION & IRB PROTOCOL

NeuroScan CDS operates as an on-premise workstation with PACS window/level presets, 3D Multi-Planar Reconstruction (MPR), and pre-operative neurosurgical planning metrics (hemisphere, subsite, midline shift). The platform includes an automated Institutional Review Board (IRB) protocol generator outlining a prospective N = 100 reader study for tertiary hospital ethics review.

VII. CONCLUSION

NeuroScan CDS bridges the gap between academic machine learning benchmarks and clinical medical reality. Through rigorous data auditing, 3D Multi-Planar Reconstruction, out-of-distribution safeguards, and transparent surgical planning, the platform delivers an auditable foundation for clinical adoption.

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